

## **Sid Mathur's Thematic ETF Screener – Relative-Strength Rotation**

2026-06-09

|                                      |           |
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## Action Center

**As of 09-Jun-2026:** holding **24 ETFs** across **6 sectors** (of **4** currently in relative-LONG). Gross exposure **100%**, cash **0%**.

Full-period: **CAGR 12.3%**, **Sharpe 0.92**, **Sortino 1.28**, **Max DD -13.6%**, Calmar **0.90**. Signal on relative strength, P&L on the absolute ETFs held; no leverage; inverse-vol weighted within the selected leader set.

Rebalance is **monthly** (last trading day decision, next-morning execution). Signals are computed daily, but positions only change at month-end; intra-month the book is static. Min-hold 30 days is honored on each per-position basis.

## Latest rebal changes

Last rebalance: **01-Jun-2026** (24 trades).

## Entered

| ticker | name                             | sector                | weight |
|--------|----------------------------------|-----------------------|--------|
| VPL    | Vanguard FTSE Pacific ETF        | Japan Equity          | 3.7%   |
| XT     | iShares Future Exponential Te... | U.S. Large Cap Equity | 5.4%   |
| XNTK   | State Street SPDR NYSE Techno... | U.S. Large Cap Growth | 4.0%   |
| MTUM   | iShares MSCI USA Momentum Fac... | U.S. Large Cap Equity | 4.2%   |
| IWC    | iShares Microcap ETF             | U.S. Mid Cap Equity   | 4.1%   |
| IWO    | iShares Russell 2000 Growth F... | U.S. Mid Cap Equity   | 4.3%   |
| VTWG   | Vanguard Russell 2000 Growth ... | U.S. Mid Cap Equity   | 4.2%   |
| FLJH   | Franklin FTSE Japan Hedged ETF   | Japan Equity          | 5.3%   |
| DBJP   | Xtrackers MSCI Japan Hedged E... | Japan Equity          | 5.2%   |

(no exits)

## Imminent moves

Next-in-line entries (eligible, in a leading sector, not yet held)

| ticker | name                             | sector                | rs_momentum |
|--------|----------------------------------|-----------------------|-------------|
| XTN    | State Street SPDR S&P Transpo... | U.S. Mid Cap Equity   | 10.9%       |
| AIQ    | Global X Artificial Intellige... | U.S. Large Cap Growth | 6.7%        |

|      |                                  |                       |      |
|------|----------------------------------|-----------------------|------|
| FV   | First Trust Dorsey Wright Foc... | U.S. Large Cap Equity | 6.7% |
| FTEC | Fidelity MSCI Information Tec... | U.S. Large Cap Growth | 6.2% |
| VGT  | Vanguard Information Tech ETF    | U.S. Large Cap Growth | 6.0% |
| IYW  | iShares U.S. Technology ETF      | U.S. Large Cap Growth | 5.7% |
| FXL  | First Trust Technology AlphaD... | U.S. Large Cap Growth | 5.6% |
| IGM  | iShares Expanded Tech Sector ... | U.S. Large Cap Growth | 5.3% |
| FTXR | First Trust Nasdaq Transporta... | U.S. Mid Cap Equity   | 4.8% |
| PDP  | Invesco Dorsey Wright Momentu... | U.S. Large Cap Equity | 4.8% |

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## Performance vs benchmarks

Table 3: Same window. Defensive Strategy = dual-momentum clusters; Market = full-universe PC1 buy-and-hold.

| Metric   | Rel-Strength | Defensive Strategy | Market |
|----------|--------------|--------------------|--------|
| CAGR     | 12.3%        | 4.7%               | 7.0%   |
| Ann vol  | 12.5%        | 7.9%               | 15.1%  |
| Sharpe   | 0.92         | 0.58               | 0.52   |
| Sortino  | 1.28         | 0.81               | 0.75   |
| Max DD   | -13.6%       | -13.4%             | -26.1% |
| Calmar   | 0.90         | 0.35               | 0.27   |
| Hit rate | 55%          | 54%                | 53%    |

Table 4: Year-by-year.

| year | start_nav | end_nav | return | sharpe | max_drawdown | n_obs |
|------|-----------|---------|--------|--------|--------------|-------|
| 2021 | 100.0     | 100.0   | 0.0%   |        | 0.0%         | 74    |
| 2022 | 100.0     | 97.7    | -2.3%  | -0.54  | -3.9%        | 251   |
| 2023 | 97.6      | 97.5    | -0.0%  | 0.02   | -8.0%        | 250   |
| 2024 | 95.8      | 110.1   | 14.9%  | 0.88   | -11.6%       | 252   |
| 2025 | 109.5     | 145.7   | 33.0%  | 2.13   | -6.3%        | 250   |
| 2026 | 146.7     | 172.6   | 17.6%  | 1.64   | -13.6%       | 108   |

## Monthly returns

Table 5: Monthly returns by year (live window).

|      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul  | Aug   | Sep   | Oct   | Nov   | Dec   |
|------|-------|-------|-------|-------|-------|-------|------|-------|-------|-------|-------|-------|
| 2021 |       |       |       |       |       |       |      |       | 0.0%  | 0.0%  | 0.0%  | 0.0%  |
| 2022 | 0.0%  | 0.0%  | 0.0%  | 0.0%  | 0.0%  | 0.0%  | 0.0% | 0.0%  | 0.0%  | 0.0%  | 0.0%  | -2.3% |
| 2023 | -0.1% | -3.3% | -0.5% | -0.0% | -1.3% | 0.6%  | 3.2% | -2.3% | -1.1% | -1.0% | -0.1% | 6.0%  |
| 2024 | -2.9% | 6.8%  | 3.4%  | -4.3% | 3.7%  | -0.0% | 1.4% | -0.4% | 2.4%  | 0.9%  | 8.5%  | -6.3% |
| 2025 | 2.3%  | 0.7%  | -0.8% | -0.1% | 0.0%  | 7.0%  | 1.3% | 6.2%  | 5.5%  | 2.4%  | 0.5%  | 3.7%  |
| 2026 | 7.4%  | 6.4%  | -8.2% | 8.0%  | 7.5%  | -2.7% |      |       |       |       |       |       |

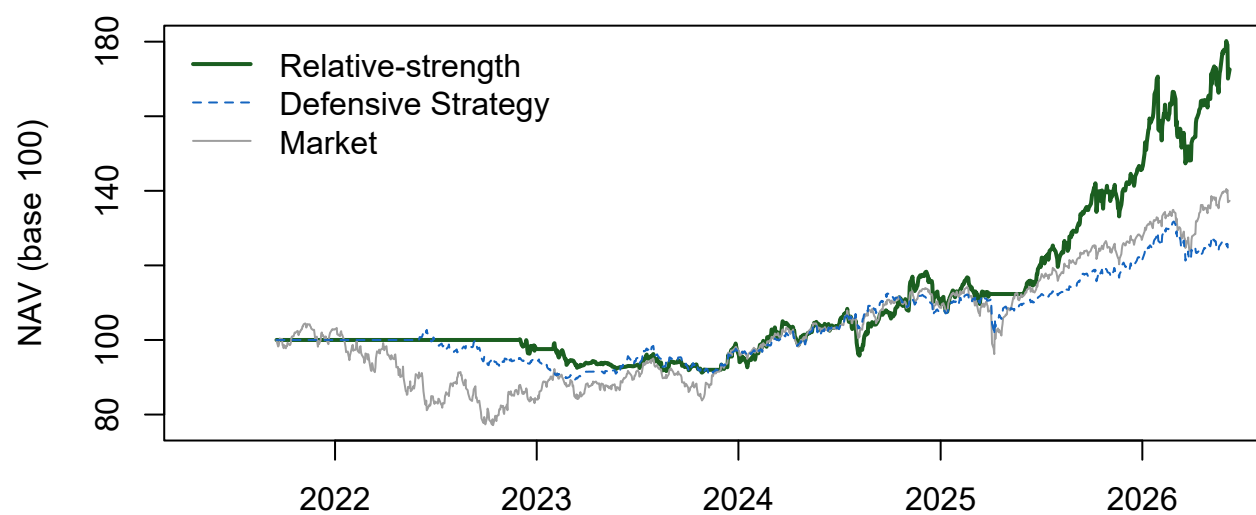


Figure 1: Growth of 100. Relative-strength rotation vs Defensive Strategy and market buy-and-hold.

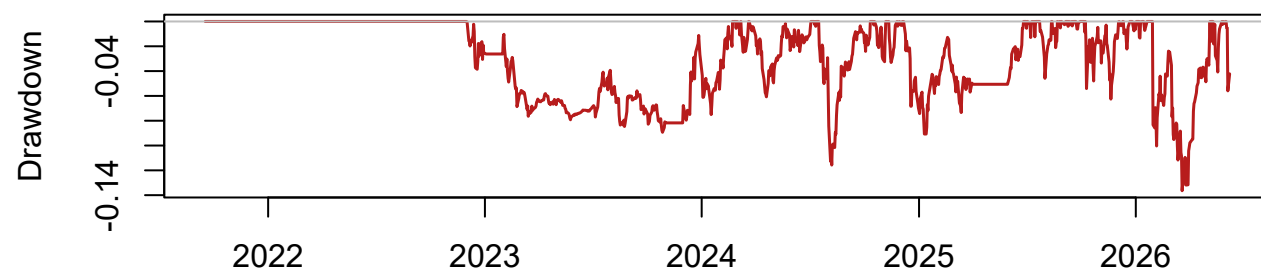


Figure 2: Drawdown of the relative-strength strategy (live window).

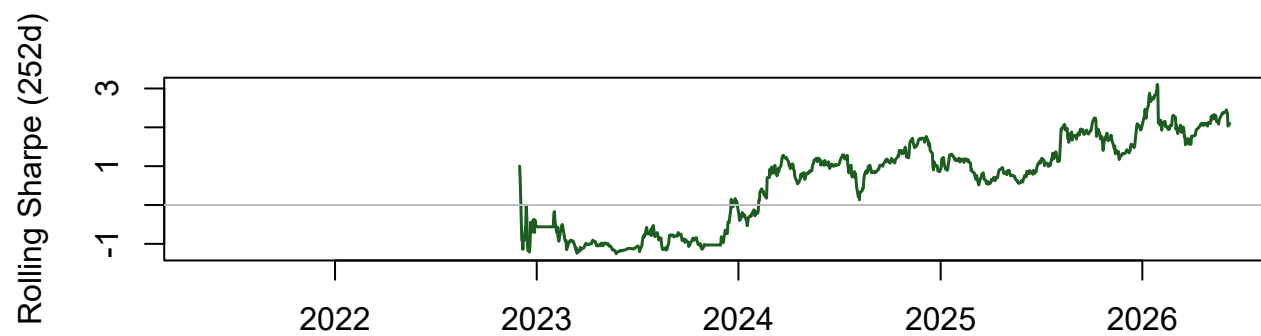


Figure 3: Rolling 252-day annualised Sharpe.



## Position Sheet

Current holdings, enriched with per-name vol, recent return, and the per-name signal state. Sorted by sector then weight desc.

Table 6: Current book. Vol-60d = trailing-60d annualised; ret-3m = 63-day cumulative; days held = calendar days in book.

| ticker | name                           | sector                  | weight | vol_60d | ret_3m | days_held | etf_rs | sector_rs |
|--------|--------------------------------|-------------------------|--------|---------|--------|-----------|--------|-----------|
| IDV    | iShares International Selec... | Developed ex-US Equity  | 6.5%   | 16.3%   | 3.7%   | 38        | FLAT   | FLAT      |
| FNDF   | Schwab Fundamental Internat... | Developed ex-US Equity  | 5.2%   | 21.4%   | 7.2%   | 38        | LONG   | FLAT      |
| PXF    | Invesco RAFI Developed Mark... | Developed ex-US Equity  | 5.0%   | 22.1%   | 7.1%   | 38        | LONG   | FLAT      |
| AAXJ   | iShares MSCI All Country As... | Emerging Markets Equity | 3.6%   | 33.1%   | 12.8%  | 38        | LONG   | FLAT      |
| AIA    | iShares Asia 50 ETF            | Emerging Markets Equity | 3.0%   | 40.6%   | 21.7%  | 38        | LONG   | FLAT      |
| EWY    | iShares MSCI South Korea ETF   | Emerging Markets Equity | 1.6%   | 70.4%   | 33.8%  | 38        | LONG   | FLAT      |
| FLJH   | Franklin FTSE Japan Hedged ... | Japan Equity            | 5.3%   | 20.0%   | 7.9%   | 7         | LONG   | LONG      |
| DBJP   | Xtrackers MSCI Japan Hedged... | Japan Equity            | 5.2%   | 20.9%   | 8.4%   | 7         | LONG   | LONG      |
| VPL    | Vanguard FTSE Pacific ETF      | Japan Equity            | 3.7%   | 30.7%   | 9.4%   | 7         | LONG   | LONG      |
| XT     | iShares Future Exponential ... | U.S. Large Cap Equity   | 5.4%   | 21.0%   | 13.6%  | 7         | LONG   | LONG      |
| MTUM   | iShares MSCI USA Momentum F... | U.S. Large Cap Equity   | 4.2%   | 28.5%   | 25.7%  | 7         | LONG   | LONG      |
| RPG    | Invesco S&P 500 Pure Growth... | U.S. Large Cap Equity   | 4.0%   | 27.9%   | 19.1%  | 38        | LONG   | LONG      |
| DWAS   | Invesco Dorsey Wright Small... | U.S. Large Cap Equity   | 4.0%   | 27.0%   | 11.8%  | 38        | LONG   | LONG      |
| ROBO   | ROBO Global Robotics and Au... | U.S. Large Cap Equity   | 3.5%   | 32.7%   | 14.8%  | 38        | LONG   | LONG      |
| XNTK   | State Street SPDR NYSE Tech... | U.S. Large Cap Growth   | 4.0%   | 32.2%   | 33.6%  | 7         | LONG   | LONG      |
| IXN    | iShares Global Tech ETF        | U.S. Large Cap Growth   | 3.8%   | 32.4%   | 29.5%  | 38        | LONG   | LONG      |
| QTUM   | Defiance Quantum ETF           | U.S. Large Cap Growth   | 3.4%   | 36.5%   | 38.8%  | 38        | LONG   | LONG      |
| XSD    | State Street SPDR S&P Semic... | U.S. Large Cap Growth   | 2.4%   | 52.6%   | 73.5%  | 38        | LONG   | LONG      |
| ISMD   | Inspire Small/Mid Cap ETF      | U.S. Mid Cap Equity     | 5.6%   | 19.0%   | 16.6%  | 38        | LONG   | LONG      |
| XSMO   | Invesco S&P SmallCap Moment... | U.S. Mid Cap Equity     | 4.5%   | 23.6%   | 12.3%  | 38        | FLAT   | LONG      |
| IWO    | iShares Russell 2000 Growth... | U.S. Mid Cap Equity     | 4.3%   | 25.4%   | 12.8%  | 7         | FLAT   | LONG      |
| VTWG   | Vanguard Russell 2000 Growt... | U.S. Mid Cap Equity     | 4.2%   | 25.5%   | 13.0%  | 7         | FLAT   | LONG      |
| IWC    | iShares Microcap ETF           | U.S. Mid Cap Equity     | 4.1%   | 28.1%   | 10.0%  | 7         | FLAT   | LONG      |
| PKB    | Invesco Building & Construc... | U.S. Mid Cap Equity     | 3.7%   | 27.4%   | 1.7%   | 38        | FLAT   | LONG      |

## Sector Overview

Table 7: All clusters, LONG first then FLAT; within each, RS-momentum desc. Held = positions taken; Members = ETFs in the cluster.

| sector                           | rs_state | rs_momentum | n_held | n_members |
|----------------------------------|----------|-------------|--------|-----------|
| U.S. Large Cap Growth            | LONG     | 14.2%       | 4      | 56        |
| U.S. Mid Cap Equity              | LONG     | 2.9%        | 6      | 68        |
| U.S. Large Cap Equity            | LONG     | 2.3%        | 5      | 80        |
| Japan Equity                     | LONG     | 2.1%        | 3      | 9         |
| Emerging Markets Equity          | FLAT     | 4.3%        | 3      | 25        |
| Broad Commodities                | FLAT     | 3.9%        | 0      | 10        |
| Financials                       | FLAT     | -0.1%       | 0      | 14        |
| Developed ex-US Equity           | FLAT     | -0.9%       | 3      | 61        |
| U.S. Large Cap Value             | FLAT     | -1.5%       | 0      | 85        |
| Real Estate                      | FLAT     | -2.4%       | 0      | 14        |
| Healthcare                       | FLAT     | -4.3%       | 0      | 17        |
| Bank Loan / Floating Rate        | FLAT     | -4.3%       | 0      | 9         |
| Materials                        | FLAT     | -4.8%       | 0      | 21        |
| Cash / Ultra-Short Fixed Income  | FLAT     | -5.2%       | 0      | 9         |
| High Yield Bond                  | FLAT     | -5.4%       | 0      | 21        |
| Municipal Bond                   | FLAT     | -5.5%       | 0      | 13        |
| Aggregate Fixed Income           | FLAT     | -6.7%       | 0      | 43        |
| Inflation-Protected Fixed Income | FLAT     | -6.8%       | 0      | 10        |
| International Fixed Income       | FLAT     | -7.1%       | 0      | 6         |
| Asia-Pacific Equity              | FLAT     | -7.7%       | 0      | 23        |
| Fixed Income                     | FLAT     | -8.1%       | 0      | 15        |
| China Equity                     | FLAT     | -9.0%       | 0      | 15        |
| Consumer Staples                 | FLAT     | -9.5%       | 0      | 11        |
| India Equity                     | FLAT     | -20.6%      | 0      | 11        |
| Gold                             | FLAT     | -25.8%      | 0      | 12        |
| Leveraged / Inverse              | FLAT     | -28.7%      | 0      | 2         |

Leading Cluster Details

Sector 5: U.S. Large Cap Growth

- Relative-strength state: **LONG** RS momentum (63d): **14.2%** held: **4 / 56** members
- Theme: U.S. Large Cap Growth



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| XNTK   | State Street SPDR NYSE Tech... | 4.0%   | 11.7%  | LONG   | 7         |
| IXN    | iShares Global Tech ETF        | 3.8%   | 8.3%   | LONG   | 38        |
| QTUM   | Defiance Quantum ETF           | 3.4%   | 16.6%  | LONG   | 38        |
| XSD    | State Street SPDR S&P Semic... | 2.4%   | 46.5%  | LONG   | 38        |

Strategy-1 replication basket (context):

| ticker | name                        | rep_wt | held_now |
|--------|-----------------------------|--------|----------|
| QQQ    | Invesco QQQ Trust, Series 1 | 103.6% |          |

### Sector 3: U.S. Mid Cap Equity

- Relative-strength state: **LONG** RS momentum (63d): **2.9%** held: **6 / 68** members
- Theme: U.S. Mid Cap Equity



#### Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| ISMD   | Inspire Small/Mid Cap ETF      | 5.6%   | 6.8%   | LONG   | 38        |
| XSMO   | Invesco S&P SmallCap Moment... | 4.5%   | 3.4%   | FLAT   | 38        |
| IWO    | iShares Russell 2000 Growth... | 4.3%   | 4.3%   | FLAT   | 7         |
| VTWG   | Vanguard Russell 2000 Growt... | 4.2%   | 4.2%   | FLAT   | 7         |
| IWC    | iShares Microcap ETF           | 4.1%   | 2.6%   | FLAT   | 7         |
| PKB    | Invesco Building & Construc... | 3.7%   | -6.1%  | FLAT   | 38        |

#### Strategy-1 replication basket (context):

| ticker | name                         | rep_wt | held_now |
|--------|------------------------------|--------|----------|
| IJH    | iShares Core S&P Mid-Cap ETF | 104.6% |          |

Sector 2: U.S. Large Cap Equity

- Relative-strength state: **LONG** RS momentum (63d): **2.3%** held: **5 / 80** members
- Theme: U.S. Large Cap Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| XT     | iShares Future Exponential ... | 5.4%   | 5.6%   | LONG   | 7         |
| MTUM   | iShares MSCI USA Momentum F... | 4.2%   | 17.6%  | LONG   | 7         |
| RPG    | Invesco S&P 500 Pure Growth... | 4.0%   | 11.2%  | LONG   | 38        |
| DWAS   | Invesco Dorsey Wright Small... | 4.0%   | 4.6%   | LONG   | 38        |
| ROBO   | ROBO Global Robotics and Au... | 3.5%   | 7.2%   | LONG   | 38        |

Strategy-1 replication basket (context):

| ticker | name                 | rep_wt | held_now |
|--------|----------------------|--------|----------|
| V00    | Vanguard S&P 500 ETF | 100.9% |          |

Sector 22: Japan Equity

- Relative-strength state: **LONG** RS momentum (63d): **2.1%** held: **3 / 9** members
- Theme: Japan Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| FLJH   | Franklin FTSE Japan Hedged ... | 5.3%   | 2.0%   | LONG   | 7         |
| DBJP   | Xtrackers MSCI Japan Hedged... | 5.2%   | 1.7%   | LONG   | 7         |
| VPL    | Vanguard FTSE Pacific ETF      | 3.7%   | 2.5%   | LONG   | 7         |

Strategy-1 replication basket (context):

| ticker | name                           | rep_wt | held_now |
|--------|--------------------------------|--------|----------|
| EWJ    | iShares MSCI Japan Index Fu... | 90.9%  |          |

Sector 7: Emerging Markets Equity

- Relative-strength state: **FLAT** RS momentum (63d): **4.3%** held: **3 / 25** members
- Theme: Emerging Markets Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| AAXJ   | iShares MSCI All Country As... | 3.6%   | 3.6%   | LONG   | 38        |
| AIA    | iShares Asia 50 ETF            | 3.0%   | 12.7%  | LONG   | 38        |
| EWY    | iShares MSCI South Korea ETF   | 1.6%   | 22.0%  | LONG   | 38        |

Strategy-1 replication basket (context):

| ticker | name                           | rep_wt | held_now |
|--------|--------------------------------|--------|----------|
| IEMG   | iShares Core MSCI Emerging ... | 92.6%  |          |

Sector 4: Developed ex-US Equity

- Relative-strength state: **FLAT** RS momentum (63d): **-0.9%** held: **3 / 61** members
- Theme: Developed ex-US Equity



Currently held from this sector:

| ticker | name                           | weight | rs_mom | etf_rs | days_held |
|--------|--------------------------------|--------|--------|--------|-----------|
| IDV    | iShares International Selec... | 6.5%   | -0.9%  | FLAT   | 38        |
| FNDF   | Schwab Fundamental Internat... | 5.2%   | 2.9%   | LONG   | 38        |
| PXF    | Invesco RAFI Developed Mark... | 5.0%   | 2.8%   | LONG   | 38        |

Strategy-1 replication basket (context):

| ticker | name                           | rep_wt | held_now |
|--------|--------------------------------|--------|----------|
| VEA    | Vanguard FTSE Developed Mar... | 96.8%  |          |



## Closed Trades

The strategy rebalances **monthly** – signals are computed daily, but positions only change at month-end (last trading day decision, execution next morning). The closed-trade ledger below is derived from the *engine's actual book*: each row is a position that the book opened and later closed, with hold duration measured from when the name first appeared in the weights to when it dropped out.

Closed positions: **359**. Mean hold: **43 days**. Median hold: **30 days**. Mean gross return: **1.8%** (absolute price over the held window).

Win rate: **56%**.

Table 20: Top 10 winning closed positions (engine-derived).

| ticker | entry      | exit       | hold_days | ret   |
|--------|------------|------------|-----------|-------|
| COPX   | 2025-12-01 | 2026-02-27 | 88        | 47.2% |
| CNXT   | 2025-08-01 | 2025-10-31 | 91        | 37.7% |
| GDXJ   | 2025-07-01 | 2025-10-31 | 122       | 34.6% |
| GDX    | 2025-08-01 | 2025-10-31 | 91        | 34.5% |
| ARKQ   | 2025-08-01 | 2025-10-31 | 91        | 29.9% |
| ARKW   | 2025-06-02 | 2025-08-29 | 88        | 27.2% |
| SIL    | 2025-07-01 | 2025-08-29 | 59        | 19.4% |
| BLOK   | 2025-06-02 | 2025-07-31 | 59        | 19.3% |
| SMH    | 2024-02-01 | 2024-03-28 | 56        | 19.0% |
| KBWB   | 2025-06-02 | 2025-08-29 | 88        | 18.7% |

Table 21: Top 10 losing closed positions (engine-derived).

| ticker | entry      | exit       | hold_days | ret    |
|--------|------------|------------|-----------|--------|
| BLOK   | 2024-03-01 | 2024-04-30 | 60        | -11.1% |
| URA    | 2026-02-02 | 2026-03-31 | 57        | -11.4% |
| GDX    | 2023-12-01 | 2024-01-31 | 61        | -11.4% |
| RFV    | 2023-02-01 | 2023-03-31 | 58        | -12.4% |
| RZV    | 2023-02-01 | 2023-03-31 | 58        | -12.9% |
| GDXJ   | 2023-12-01 | 2024-01-31 | 61        | -14.5% |
| EZA    | 2026-03-02 | 2026-03-31 | 29        | -16.2% |
| PSI    | 2023-09-01 | 2023-10-31 | 60        | -19.2% |
| GDX    | 2026-03-02 | 2026-03-31 | 29        | -21.9% |
| GOAU   | 2026-03-02 | 2026-03-31 | 29        | -22.3% |

## ETF RS Leaders &amp; Laggards

The relative-strength analogue of strategy 1's Net Signal Strength: ETFs ranked by their **63-day relative-strength momentum** (ETF vs its own cluster PC1). Bullish list is restricted to ETFs in clusters currently in relative-LONG; bearish list is unrestricted.

Table 22: Top 20 most bullish ETFs by RS momentum (restricted to leading sectors).

| ticker | name                             | sector                | rs_mom | held | etf_rs |
|--------|----------------------------------|-----------------------|--------|------|--------|
| XSD    | State Street SPDR S&P Semicon... | U.S. Large Cap Growth | 46.5%  | Y    | LONG   |
| FTXL   | First Trust Nasdaq Semiconduc... | U.S. Large Cap Growth | 41.8%  |      | FLAT   |
| SOXX   | iShares PHLX SOX Semiconducto... | U.S. Large Cap Growth | 38.7%  |      | FLAT   |
| PSI    | Invesco Semiconductors ETF       | U.S. Large Cap Growth | 33.6%  |      | FLAT   |
| SMH    | VanEck Semiconductor ETF         | U.S. Large Cap Growth | 24.5%  |      | FLAT   |
| MTUM   | iShares MSCI USA Momentum Fac... | U.S. Large Cap Equity | 17.6%  | Y    | LONG   |
| QTUM   | Defiance Quantum ETF             | U.S. Large Cap Growth | 16.6%  | Y    | LONG   |
| PTF    | Invesco Dorsey Wright Technol... | U.S. Large Cap Growth | 15.6%  |      | FLAT   |
| QTEC   | First Trust NASDAQ-100-Techno... | U.S. Large Cap Growth | 14.5%  |      | FLAT   |
| PSCT   | Invesco S&P SmallCap Informat... | U.S. Large Cap Growth | 13.2%  |      | FLAT   |
| SPHB   | Invesco S&P 500 High Beta ETF    | U.S. Large Cap Equity | 12.0%  |      | FLAT   |
| XNTK   | State Street SPDR NYSE Techno... | U.S. Large Cap Growth | 11.7%  | Y    | LONG   |
| RPG    | Invesco S&P 500 Pure Growth E... | U.S. Large Cap Equity | 11.2%  | Y    | LONG   |
| XTN    | State Street SPDR S&P Transpo... | U.S. Mid Cap Equity   | 10.9%  |      | LONG   |
| NXTG   | First Trust Indxx NextG ETF      | U.S. Large Cap Growth | 10.7%  |      | FLAT   |
| PRN    | Invesco Dorsey Wright Industr... | U.S. Large Cap Equity | 9.6%   |      | FLAT   |
| XLK    | State Street Technology Selec... | U.S. Large Cap Growth | 9.0%   |      | FLAT   |
| FPXI   | First Trust International Equ... | U.S. Large Cap Equity | 8.3%   |      | FLAT   |
| IXN    | iShares Global Tech ETF          | U.S. Large Cap Growth | 8.3%   | Y    | LONG   |
| ROBO   | ROBO Global Robotics and Auto... | U.S. Large Cap Equity | 7.2%   | Y    | LONG   |

Table 23: Top 20 most bearish ETFs by RS momentum (full universe).

| ticker | name                             | sector                | rs_mom  | sector_rs |
|--------|----------------------------------|-----------------------|---------|-----------|
| PPLT   | abrdn Physical Platinum Share... | India Equity          | -211.0% | FLAT      |
| NOBL   | ProShares S&P 500 Dividend Ar... | India Equity          | -61.2%  | FLAT      |
| SQQQ   | ProShares UltraPro Short QQQ     | Leveraged / Inverse   | -24.9%  | FLAT      |
| IPAY   | Amplify Digital Payments ETF     | U.S. Large Cap Growth | -22.5%  | LONG      |
| EIDO   | iShares MSCI Indonesia ETF       | India Equity          | -22.4%  | FLAT      |
| FINX   | Global X FinTech ETF             | U.S. Large Cap Growth | -22.1%  | LONG      |
| PNQI   | Invesco Nasdaq Internet ETF      | U.S. Large Cap Growth | -19.4%  | LONG      |
| IBUY   | Amplify Online Retail ETF        | U.S. Large Cap Growth | -17.7%  | LONG      |
| ARKK   | ARK Innovation ETF               | U.S. Large Cap Growth | -16.1%  | LONG      |
| IHI    | iShares U.S. Medical Devices ... | Healthcare            | -14.4%  | FLAT      |

|      |                                  |                         |        |      |
|------|----------------------------------|-------------------------|--------|------|
| QYLD | Global X NASDAQ 100 Covered C... | U.S. Large Cap Growth   | -14.3% | LONG |
| PTLC | Pacer Trendpilot US Large Cap... | U.S. Large Cap Growth   | -14.2% | LONG |
| PPA  | Invesco Aerospace & Defense E... | U.S. Large Cap Equity   | -13.5% | LONG |
| ITA  | iShares U.S. Aerospace & Defe... | U.S. Large Cap Equity   | -13.4% | LONG |
| ITB  | iShares U.S. Home Constructio... | U.S. Mid Cap Equity     | -13.2% | LONG |
| SDIV | Global X SuperDividend ETF       | Emerging Markets Equity | -12.9% | FLAT |
| DBA  | Invesco DB Agriculture Fund      | Broad Commodities       | -12.6% | FLAT |
| URA  | Global X Uranium ETF             | Materials               | -12.5% | FLAT |
| XLC  | State Street Communication Se... | U.S. Large Cap Equity   | -12.5% | LONG |
| ESPO | VanEck Video Gaming and eSpor... | U.S. Large Cap Equity   | -12.3% | LONG |

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## Watchlist

Table 24: Eligible leaders in leading sectors not currently held (next in line at next rebal).

| ticker | name                             | sector                | rs_mom |
|--------|----------------------------------|-----------------------|--------|
| XTN    | State Street SPDR S&P Transpo... | U.S. Mid Cap Equity   | 10.9%  |
| AIQ    | Global X Artificial Intellige... | U.S. Large Cap Growth | 6.7%   |
| FV     | First Trust Dorsey Wright Foc... | U.S. Large Cap Equity | 6.7%   |
| FTEC   | Fidelity MSCI Information Tec... | U.S. Large Cap Growth | 6.2%   |
| VGT    | Vanguard Information Tech ETF    | U.S. Large Cap Growth | 6.0%   |
| IYW    | iShares U.S. Technology ETF      | U.S. Large Cap Growth | 5.7%   |
| FXL    | First Trust Technology AlphaD... | U.S. Large Cap Growth | 5.6%   |
| IGM    | iShares Expanded Tech Sector ... | U.S. Large Cap Growth | 5.3%   |
| FTXR   | First Trust Nasdaq Transporta... | U.S. Mid Cap Equity   | 4.8%   |
| PDP    | Invesco Dorsey Wright Momentu... | U.S. Large Cap Equity | 4.8%   |
| XMMO   | Invesco S&P MidCap Momentum E... | U.S. Large Cap Equity | 4.1%   |
| RWJ    | Invesco S&P SmallCap 600 Reve... | U.S. Mid Cap Equity   | 3.8%   |
| RZV    | Invesco S&P SmallCap 600 Pure... | U.S. Mid Cap Equity   | 3.5%   |
| HEWJ   | iShares Currency Hedged MSCI ... | Japan Equity          | 2.1%   |
| QMOM   | Alpha Architect U.S. Quantita... | U.S. Large Cap Equity | 2.1%   |

## Run Parameters

| parameter                       | value        |
|---------------------------------|--------------|
| Holding mode                    | etf_topk     |
| Signal (short/buy/sell MA)      | 5 / 50 / 20  |
| Sell cushion k                  | 0.5          |
| Min hold (days)                 | 30           |
| Rearm blocked buys              | TRUE         |
| RS ranking window               | 63           |
| Leading clusters (n)            | 6            |
| Leaders per cluster (top_k)     | 3            |
| Weight scheme                   | inverse_vol  |
| Vol window (for inverse-vol)    | 60           |
| Per-name cap                    | 20%          |
| Relative entry filter (abs-mom) | 252d         |
| Absolute overlay (abs-mom)      | 252d         |
| Market gate                     | ON (200d MA) |
| Rebalance                       | monthly      |
| Cost (bps)                      | 10           |